

Shape-aware normative modelling reveals distinct cortical developmental pathways

Yourong Guo^{1,2}, Alexander Hammers^{1,3}, & Emma C. Robinson^{1,2,5}, Jonathan O'Muircheartaigh^{2,4,5}

¹Research Department of Biomedical Computing, School of Biomedical Engineering and Imaging Sciences, King's College London

²Centre for the Developing Brain, King's College London

³King's College London and Guy's and St. Thomas' PET Centre, King's College London

⁴Department of Forensic and Neurodevelopmental Sciences, King's College London

⁵MRC Centre for Neurodevelopmental Disorders, King's College London

Introduction:

Over recent years, normative models [1] have become a powerful framework for distinguishing disease-related deviations from healthy variation by modelling 'normative' age-dependent trajectories from large cohorts of healthy individuals. However, their accuracy depends critically on how well cortical anatomy is aligned across subjects. Conventional diffeomorphic image registration to a single population average template cannot perfectly align sulci and gyri across individuals, leading to blending of distinct folding tissues and blurring underlying developmental signal.

This limitation has important clinical implications. Many neurodevelopmental and neurological conditions manifest differently across the folding tissues. For example, in epilepsy, cortical malformation such as focal cortical dysplasia often arise in sulcal fundi and are accompanied by local thickness variabilities. Because cortical thickness varies systematically with folding position – thicker in gyral crowns and thinner in sulcal pits – such abnormalities can be obscured when folding pattern are misaligned. Although computational detection tools have shown promise, their performance varies markedly across the cortex: false positives and negatives increase in regions with high folding variability [2].

These challenges highlight the need for normative frameworks that can account for typical anatomical shape variability. In previous work, we demonstrated that major cortical folding variants can be characterised and aligned using hierarchical surface registration (MSM-HT) [3], which groups subjects with similar folding patterns, and progressively aligns them through intermediate templates to a common space. Here, we apply MSM-HT to developmental data from the Lifespan Human Connectome Project-Development (HCP-D) [4], enabling a shape-aware normative framework that preserves the anatomical distinction between gyral crowns, sulcal banks, and sulcal fundi. This allows vertex-wise developmental trajectories to be modelled within biologically meaningful tissue components, providing a more precise reference for detecting atypical or disease-related patterns.

Methods:

Data from 652 subjects aged 5 to 21 years (351 females) from HCP-D were used to build vertex-wise normative models of cortical thickness. For demonstration purposes, analyses were restricted to the frontal lobe of the left hemisphere. First, we applied MSM-HT to the dataset via a template matching procedure. Each hemisphere was registered to all 30 frontal lobe templates [3] using ultra-fast learning-based image registration and pairwise similarity scores computed using correlation of sulcal depth and Dice overlap of the gyral crowns. Each subject was assigned to the template with the highest similarity and then aligned through the hierarchy to a common reference space (Figure 1A).

This process reduces experimental noise from image registration, since it allows samples to be clustered – at each vertex – by tissue type (Figure 1B). This was achieved through fitting vertex-wise Mixture of Gaussian Processes (MoGP) models to the MSM-HT aligned sulcal depth maps. MoGP models the observed developmental pattern as a weighted combination of several Gaussian processes (GPs), allowing each vertex to follow one of multiple trajectory types. This produces per-subject probabilities that a vertex follows a crown, bank or fundus-like trajectory. A post-hoc Markov Random Field was then applied to enforce spatial coherence, yielding a segmentation of the cortex into tissue components for each subject.

Finally, developmental pathways for cortical thickness were modelled within each tissue component using separate GP models and compared against a single naive GP model that was fit across all subjects. Model accuracy was quantified by mean absolute error (MAE) within the same components between the predicted and observed values (in mm). Split-half cross-validation, stratified by age, was used, ensuring each subject contributed once to the test set. The final normative curves were then refitted using the full dataset.

Results:

Following MSM-HT alignment, samples are more consistently aligned – meaning that at any given location individuals tend to cluster into groups that either have a gyrus or a sulcus at each location (Figure 1B). By contrast, conventional MSM-P registration to a single population template mixes all folding variants at each location, making it impossible to separate them. Figure 1C displays the output of MoGP clustering on the MSM-HT aligned data. Three classes are modelled: gyral crowns (positive values), sulcal banks (near zero), and sulcal fundi (negative values). The resulting segmentations are shown for one subject in Figure 1C.

From these classes it then becomes possible to model distinct development trajectories for gyral crowns and sulcal fundi: In the HCP-D cohorts, cortical thickness was consistently greatest for crowns and lowest for sulcal fundi, with all components showing an age-related decrease during childhood and adolescence (Figure 2A). Component-specific GP models consistently achieve lower MAE than those of the baseline models that ignore tissue assignment (evaluated within the same components, $P < 0.0001$, Figure 2B), indicating that folding-informed segmentation improves the precision of modelling for cortical thickness.

Conclusion:

Folding variability introduces regional bias in normative models by blending distinct cortical tissues. Here we demonstrate that including anatomical shape priors substantially improves the precision of normative modelling for cortical thickness. By preserving anatomically meaningful subdivisions of the cortical sheet, this framework provides a more biologically grounded reference for typical development and may increase sensitivity to subtle deviations in clinical populations.

Figures:

Figure 1: Building shape-aware normative models that account for cortical folding variability. **A.** Addressing registration. Scheme draw of hierarchical surface registration (MSM-HT) methods. After template matching, subjects were assigned to templates with highest similarity and aligned through the hierarchy to a common space. **B.** Separating sulcal/gyral trajectories. An example vertex where MSM-HT alignment reveals a clearer separation into distinct developmental trajectories (illustrated using subjects from two templates). **C.** Characterising different trajectories using Mixture of Gaussian Processes (MoGP) Model. This produces per-subject segmentation of the cortex into different folding tissue components (example subject shown).

Figure 2: Vertex-wise normative models of cortical thickness from HCP-D cohort. **A.** For an example vertex, developmental trajectories were estimated using naive GPs fitted either to all subjects (group-fit) or to fundus, bank, or crown subsets (subset-fit). **B.** Subset models produced lower MAE values (in mm) in both gyral and sulcal regions, demonstrating that folding-based segmentation improves model accuracy.

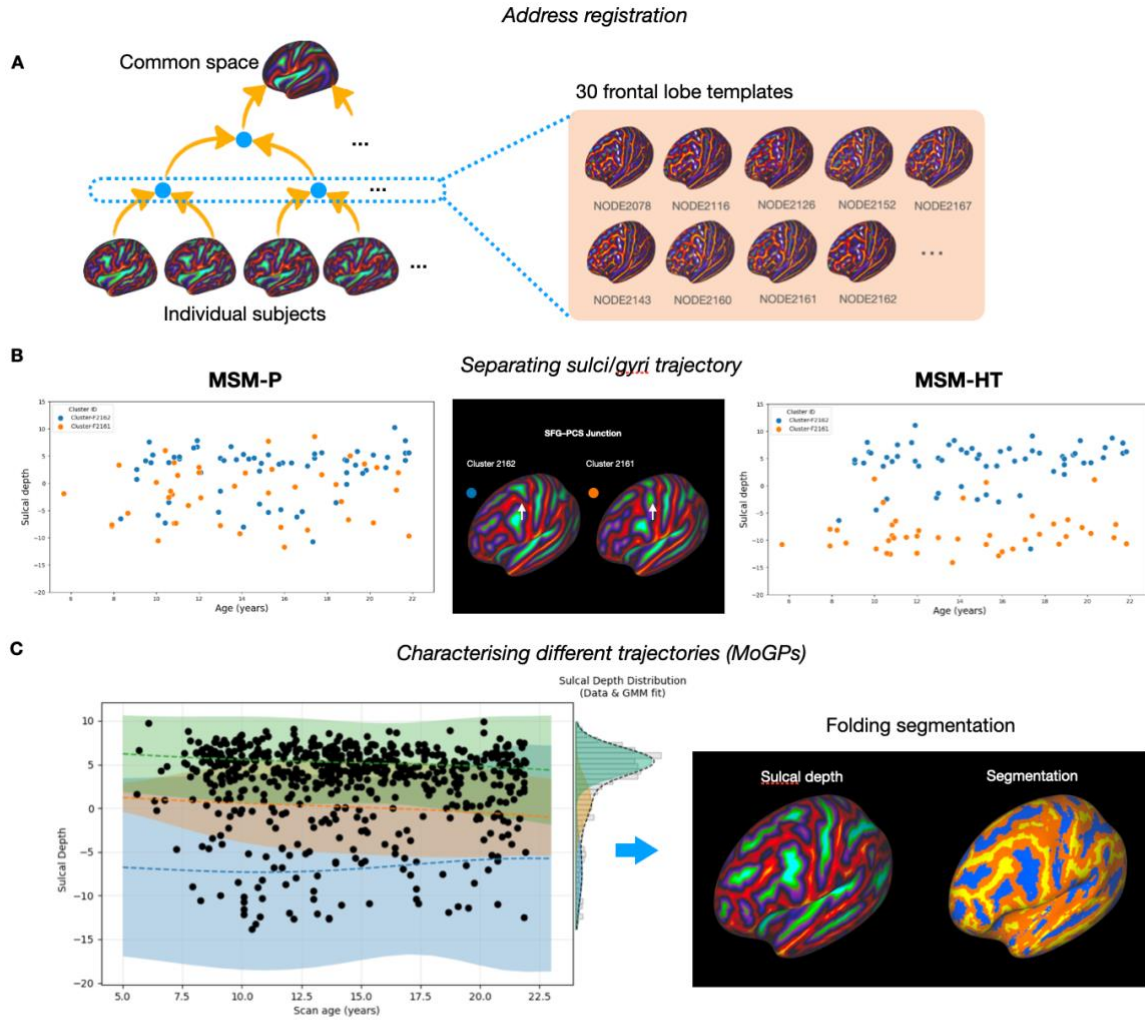


Figure 1: Building shape-aware normative models that account for cortical folding variability. **A.** Addressing registration. Scheme draw of hierarchical surface registration (MSM-HT) methods. After template matching, subjects were assigned to templates with highest similarity and aligned through the hierarchy to a common space **B.** Separating sulci/gyri trajectories. An example vertex where MSM-HT alignment reveals a clearer separation into distinct developmental trajectories, with positive values corresponding to gyra crowns and negative values to sulcal fundi. **C.** Characterising different trajectories using Mixture of Gaussian Processes (MoGPs) Model. This produces per-subject segmentation of the cortex into different folding tissue components.

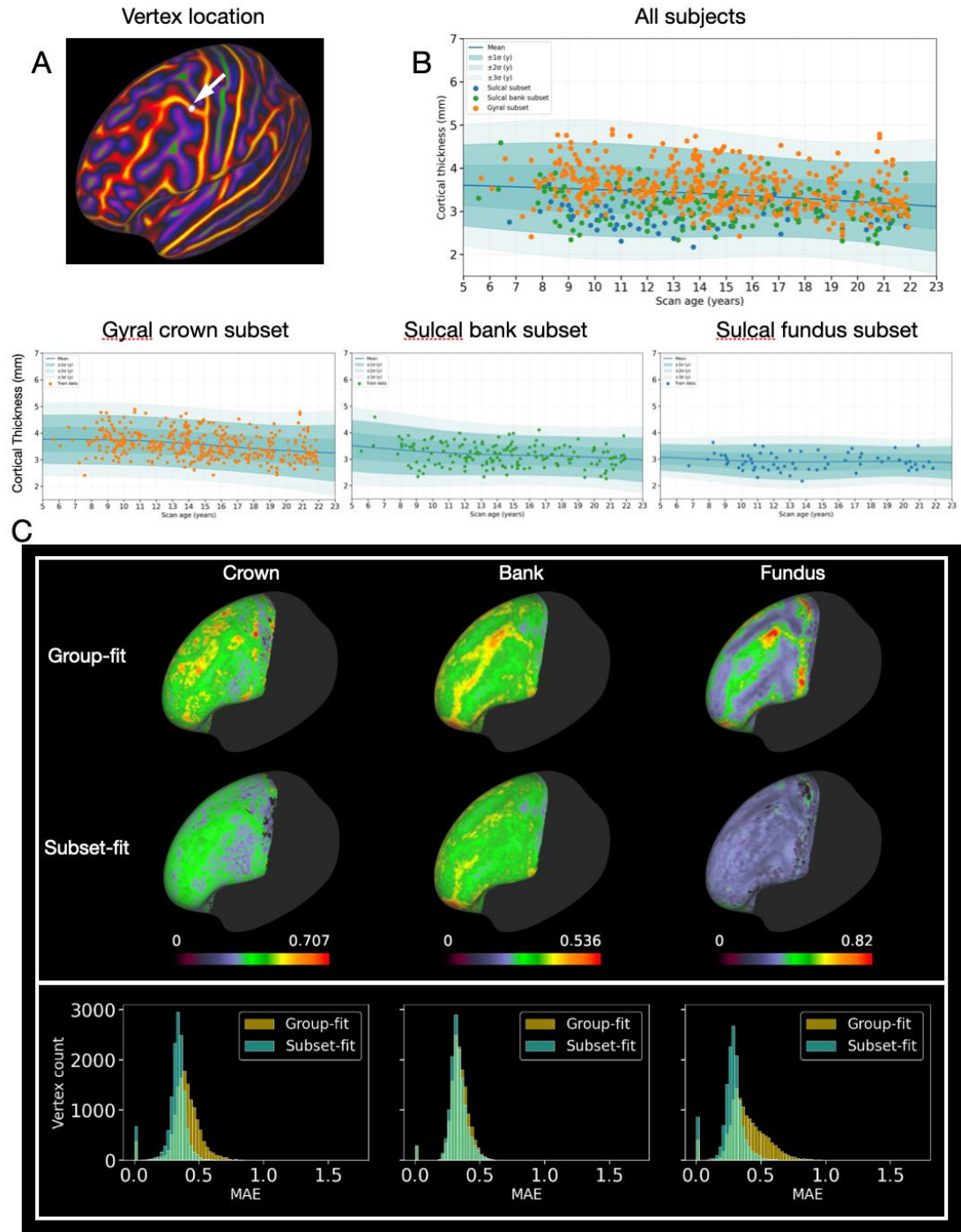


Figure 2: Vertex-wise normative models of cortical thickness from HCP-D cohort. **A.** Segmentation of brain cortex with folding tissue probability from MoGPs. **B.** At each vertex, developmental trajectories were estimated using naive GPs fitted either to all subjects (group-fit) or to fundus, bank, or crown subsets (subset-fit). **C.** Subset models produced lower MAE values (in mm) in both gyral and sulcal regions, demonstrating that folding-based segmentation improves model accuracy.

References:

1. Marquand, A. F. (2016). Understanding heterogeneity in clinical cohorts using normative models: beyond case-control studies. *Biological Psychiatry*, 80(7), 552-561.
2. Bunyamin, J., Sinclair, B., Law, M., Kwan, P., O'Brien, T. J., & Neal, A. (2025). Voxel-based and surface-based cortical morphometric MRI applications for identifying the epileptogenic zone: A narrative review. *Epilepsia Open*, 10(2), 380-397.
3. Guo, Y., Suliman, M. A., Williams, L. Z., Besenczi, R., Liang, K., Dahan, S., ... & Robinson, E. C. (2025). Motifs of brain cortical folding from birth to adulthood: structural asymmetry and folding-functional links. *bioRxiv*, 2025-03.
4. Somerville, L. H., Bookheimer, S. Y., Buckner, R. L., Burgess, G. C., Curtiss, S. W., Dapretto, M., ... & Barch, D. M. (2018). The Lifespan Human Connectome Project in Development: A large-scale study of brain connectivity development in 5–21 year olds. *Neuroimage*, 183, 456-468.